

Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years of analytical method development, multi-technique characterization, and DOE/SPC process-development experience across small-molecule, polymer, thin-film, and functional materials systems. Currently leads bio-based polymer process-development work at CU Boulder, building reproducible analytical datasets, stability/degradation readouts, SOP-ready documentation, and technical reports that support development decisions.

CORE COMPETENCIES

Analytical Method Development

Small Molecule / Peptide-Oligo Adjacent Characterization

Chromatography & MS Exposure

NMR / FT-IR / UV-Vis-NIR

Stability & Shelf-Life Assessment

DOE/SPC

GMP Documentation Mindset

Regulatory Submission Support

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided polymer process-development experiments, defining critical process parameters, analytical readouts, reproducibility criteria, and SOP-ready controls for scale-up decisions.
- Develop stability/degradation and release-relevant characterization readouts using FT-IR, UV-Vis-NIR-adjacent spectroscopy, thermal analysis, and SPC concepts to connect process conditions with material quality attributes.
- Translate process and characterization data into technical reports, troubleshooting recommendations, and documented methods suitable for cross-functional development review.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Designed and characterized 15+ organic electrochromic material variants using UV-Vis-NIR, FT-IR, cyclic voltammetry, spectroelectrochemistry, NMR, GPC, TGA, DSC, and mass-spectrometry-informed analysis.
- Assessed structure-property-stability relationships, degradation pathways, impurity-/failure-mode hypotheses, and method comparability to guide molecular and formulation/process decisions.
- Built rigorous experimental documentation and datasets supporting method development, reproducibility assessment, and root-cause analysis.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded molecular thin-film process development, linking CVD/surface preparation with morphology, interfaces, optical performance, and reproducible process controls.
- Prepared technical data packages connecting process variables, material performance, and risk/quality decisions for multidisciplinary stakeholders.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Optimized polymer thin-film deposition using DOE strategies and in-situ QCM-D process monitoring, characterizing film-growth kinetics to improve coating reproducibility.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed operations for a 15-member process-development laboratory, implementing equipment tracking, calibration routines, SOPs, and safety protocols that improved instrument uptime by 30%.
- Trained researchers on instrument workflows, experimental documentation, and safe handling while coordinating multiple concurrent materials-development projects.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Analytical Method Development and Stability Study

Chemistry of Materials 2025

Mapped degradation, redox-instability, and performance failure modes across functional material variants using orthogonal analytical characterization and documented method comparisons.

UV-Vis-NIR, FT-IR, CV, NMR, GPC, TGA/DSC, MS-informed analysis

Bio-Based Polymer Scale-Up Process

Current

Defined DOE-guided process windows, analytical readouts, reproducibility criteria, and SOP-ready specifications for scalable polymer systems.

DOE, SPC, FT-IR, UV-Vis-NIR, thermal analysis, technical reports

AFOSR Defense Thin-Film Coatings

DoD Funded

Connected CVD and surface-preparation variables to coating morphology, interfaces, and optical/material performance in thin-film systems.

CVD, QCM-D, spectroscopy, process documentation

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials

Advisor: Prof. Seth Marder

(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Analytical & Characterization: FT-IR, UV-Vis-NIR, NMR, GPC/SEC, HPLC exposure, GC-MS/LC-MS exposure, cyclic voltammetry, spectroelectrochemistry, TGA, DSC, QCM-D

Method Development & Stability: Analytical readouts, degradation studies, impurity/failure-mode hypotheses, reproducibility studies, method documentation, SOP-ready controls

Process & Quality Mindset: DOE, SPC, process windows, troubleshooting, root-cause analysis, technical reports, validation-package adjacency, tech-transfer documentation

Data & Communication: Experimental design, data visualization, Excel, Python/basic data analysis, JMP/basic, cross-functional technical presentations

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.

2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)